ML Performance Summary

Synthetic CTPA triage model—scenario evidence for Section 2

Synthetic training material—not clinical guidance. No live Microsoft Foundry access is required.

Model and test-set context

The supplied model is a synthetic binary triage classifier that assigns a score used to prioritize CTPA studies for radiologist review. It is not an autonomous diagnostic model. Test set: 1,200 synthetic studies, including 300 reference-critical and 900 reference-non-critical cases. The scenario team is considering 0.58 as the operating threshold because the intended workflow prioritizes sensitivity while retaining a minimum precision floor.

Table 1. Performance at Candidate Operating Threshold 0.58

Metric	Value	Scenario Acceptance Criterion	Meets Criterion?
Sensitivity/recall (critical class)	0.960	>= 0.95	Yes
Specificity	0.780	No fixed scenario minimum	Reference only
False-negative rate	0.040	<= 0.05	Yes
Precision/PPV (critical class)	0.593	>= 0.58	Yes
F1 score	0.733	No fixed scenario minimum	Reference only
Accuracy	0.825	No fixed scenario minimum	Reference only
ROC-AUC	0.940 (scenario-supplied 95% interval: 0.920–0.960)	No fixed scenario minimum	Reference only

Interpretation note: the score threshold and acceptance criteria above are scenario-defined. Do not compare them with an assumed universal “clinical tolerance.”

Table 2. Confusion Matrix at Threshold 0.58

	Reference Critical	Reference Non-Critical	Total
Predicted priority/critical	288	198	486
Predicted non-priority/non-critical	12	702	714
Total	300	900	1200

Table 3. Threshold Trade-Off Summary

Threshold	Sensitivity	Precision	FNR	Specificity	F1
0.45	0.980	0.505	0.020	0.680	0.667
0.58	0.960	0.593	0.040	0.780	0.733
0.70	0.910	0.684	0.090	0.860	0.781

Table 4. Image-Quality Subgroup Performance at Threshold 0.58

Image-Quality Subgroup	Critical Cases	TP	FN	Sensitivity	Scenario Minimum
Adequate contrast	250	245	5	0.980	0.900
Borderline/low contrast	50	43	7	0.860	0.900

The low-contrast subgroup does not meet the scenario-defined subgroup criterion. The workflow diagram provides a technical-quality review path that may be used as a mitigation; the learner must decide how this affects readiness.

Evidence pointers

- ML Table 1: headline metrics at threshold 0.58.
- ML Table 2: exact confusion-matrix counts.
- ML Table 3: threshold trade-offs; use these rather than inventing a new threshold.
- ML Table 4: subgroup evidence relevant to low-contrast studies.
- Scenario Guide: operating expectations and acceptance criteria.